Nguyen Ba Thinh

IoT & Artificial Intelligence Engineer

AIoT Engineer · Edge AI · Computer Vision · Embedded Systems

AIoT Engineer focused on Computer Vision and Edge AI on embedded devices. Hands-on experience deploying real-time AI models on Jetson Nano, Raspberry Pi 5 and Hailo-8L. Currently expanding into LLM, RAG and AI automation.



CONTACT

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SKILLS

Computer Vision · Edge AI · PyTorch · ONNX · TensorRT · OpenCV · Hailo-8L · Jetson Nano · Raspberry Pi · Embedded Linux · Python · C++ · Go · LLM · RAG · AI Agent · n8n · Docker

EDUCATION

Bachelor of Engineering, IoT & AI
Industrial University of Ho Chi Minh City (IUH)
2021 – 2026

EXPERIENCE

BEE HOOME TECHNOLOGY JSC

Fullstack

04/2026 – 09/2026

AI Chatbot for Facebook & Instagram Pages

- Built an end-to-end automated chatbot on n8n that replies to Facebook Messenger and Page comments 24/7 for a premium kitchen-cabinet brand.
- Integrated a RAG pipeline (PostgreSQL + PGVector, Gemini embeddings) to ground answers in the company knowledge base.
- Designed an AI sales-consultant agent (LangChain + Gemini) with a controlled Canada persona to consult products and qualify leads.
- Automated knowledge-base sync: Google Drive trigger, ~800-token chunking, embedding and upsert with no manual work.
- Implemented per-user bot ON/OFF for human handover, plus echo/duplicate filtering.
- Used PostgreSQL chat memory for multi-turn context and n8n Data Tables for per-customer message status.
- Self-hosted the stack with Docker (n8n, PostgreSQL + PGVector) on a Linux VPS.

Tech stack: n8n, Webhooks · Gemini, LangChain, RAG · PostgreSQL + PGVector · Docker Compose, Linux VPS · Facebook Graph API v19.0, Google Drive API · JavaScript, JSON

CERTIFICATES

Engineering Project (eProjects)

ASU and Dow Vietnam STEM Program

PUBLICATION

Efficient Face Recognition on Embedded Devices via Structured Pruning and Stereo-Camera Feature Fusion

Le-Chi T., Nguyen B. T., Cao V. K.

Springer CCIS, 2026, pp. 146–157

doi.org/10.1007/978-3-032-21628-1_11

EXPERIENCE

HBQ Technology

04/2025 – 04/2026

R&D

4G/5G Drone Streaming & Control System

- Developed a 4G/5G-enabled drone system on a custom embedded platform.
- Implemented real-time video & telemetry streaming via RTSP with <300 ms latency.
- Customized and integrated QGroundControl, improving operational efficiency by ~40%.
- Designed TCP/UDP communication and authentication, reducing packet loss by ~5% on unstable networks.
- Deployed computer vision for landing-zone detection with ~80% real-world accuracy.

Tech stack: Go, Python, C++ · OpenCV, GStreamer, RTSP/MediaMTX · Go REST, WebSocket, gomavlib · React, TypeScript · WireGuard, Linux · PX4, MAVLink, QGroundControl, Qt 6 · Raspberry Pi CM5, SIM7600G-H-M2

PROJECTS

Edge AI Face Recognition System

05/2025 – 01/2026

- Built an edge face recognition system on Raspberry Pi 5 with the Hailo-8L AI accelerator.
- Achieved ~25 FPS real-time inference under edge constraints.
- Applied quantization and pruning, reducing model size by ~30% and video processing time to ~3 s.
- Developed custom inference modules, improving recognition accuracy by ~6%.
- Improved robustness with pre-processing and post-processing techniques.

Tech stack: Python, QML, Shell · PyTorch, SCRFD, ONNX · Hailo Runtime, hailo_platform SDK · OpenCV, Picamera2, V4L2 · Raspberry Pi 5, Hailo-8L M.2, dual OV5647

Personal Project: Muscle Group Detection from Video

- Converted and optimized models through a PyTorch → ONNX → TensorRT pipeline.
- Reduced inference latency by ~30% through TensorRT optimization.
- Built deployment pipelines for scalable edge/embedded AI systems.

Tech stack: Python · PyTorch, ONNX, TensorRT · OpenCV, YOLO · NVIDIA Jetson Nano B01, OV5647